
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

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Abstract

1 Forecast accuracy does not by itself establish scientific forecasting ability. We
2 test five conditions that connect a score to a scientific claim: held-out generaliza-
3 tion, complete failure accounting, matched intervention, structural identifiability,
4 and stability across systems. We apply these tests to k -pendulum ($k \in \{1, 2, 3\}$)
5 and Lorenz-63 forecasts from language models, a time-series foundation model,
6 learned dynamics, sparse identification, and classical integrators. Each test changes
7 a substantive conclusion. Learned-model error rises from about 0.26 rad on training
8 trajectories to 0.97–1.14 rad on held-out trajectories, where linearized dynamics
9 and nearest-neighbor lookup perform better. Retaining failed forecasts lowers the
10 reported performance of models with incomplete coverage. An unmatched compar-
11 ison says that hiding physical constants helps; a paired intervention instead finds a
12 $+0.069$ rad penalty. Two exact scale symmetries prove that the absolute constants
13 are not identifiable from angle trajectories. Increasing the Neural ODE training
14 set improves error from 1.090 to 0.861 rad but does not reach known-equation
15 integration. The pendulum ordering also fails to transfer to Lorenz-63, where
16 SINDy recovers the polynomial dynamics to numerical precision. These results
17 show that a correct leaderboard can still support incorrect scientific conclusions.
18 We release the prompts, outputs, checkpoints, seeds, and compute records needed
19 to repeat the audit.

20 1 Introduction

21 Scientific claims require more than low forecast error. A result may depend on training trajectories,
22 dropped failures, unmatched inputs, an unidentifiable target, or one favorable system. We study these
23 failure modes directly. Given state $\mathbf{s}_t$, the task is to predict $\mathbf{s}_{t+T}$. The k -link pendulum is integrable
24 at $k=1$ and chaotic for $k \geq 2$, has high-precision ground truth, and permits controlled changes to
25 physical metadata. Lorenz-63 provides a second system with a different observation model and
26 hypothesis class.

27 Our contribution is a claim audit for dynamical-system forecasts. It tests held-out generalization,
28 counts every failed forecast, records the information available to each method, changes disclosure
29 on identical trajectories, checks identifiability before scoring parameter recovery, varies the training
30 budget, and tests whether conclusions transfer to a second system. Table 1 states the conclusion
31 before and after each test. The central result is empirical and direct: all five tests alter or reject a
32 conclusion supported by the unaudited comparison.

Table 1: Conclusions before and after the claim audit. Each row changes one part of the evaluation while preserving the relevant model outputs or comparison set.

Test	Unaudited conclusion	Audited conclusion
Generalization	Learned dynamics forecast accurately.	Error rises about fourfold; simple baselines perform better.
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Disclosure	Withholding constants improves forecasts.	Withholding constants increases pooled error by 0.069 rad.
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler-Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models such as Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024] are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We evaluate these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Here the output is a numerical state at a specified time in a chaotic trajectory.

Data-driven identification and abstention. SINDy fits sparse governing equations from measured trajectories [Brunton et al., 2016]; nonlinear vector autoregression provides a compact reservoir-computing formulation [Gauthier et al., 2021]. Structural identifiability asks which parameter combinations can be recovered from observed outputs, including under parameter symmetries [Borgqvist et al., 2024]. Selective prediction ordinarily reports risk together with coverage [Geifman and El-Yaniv, 2017]. Here failed cells remain in the main score because models are compared at fixed task coverage.

3 Benchmark Design

Choice of systems. We use the k -link pendulum to vary the number of coupled links while retaining a high-precision numerical solution. Its constants can also be hidden without changing the underlying trajectory, which permits the matched test in §5.2. The smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system. We did not rerun the LLM or disclosure experiments on Lorenz.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the

Table 2: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 3: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

68 RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference
69 by construction, not an independent competitor. Euler and the symplectic integrator are distinct
70 known-physics methods. Inputs available to each family are stated in §3.3.

71 3.2 Regimes and grid

72 Three regimes test generalization to non-standard constants (Table 2). In `changed_hidden` the
73 constants (g , L , m , damping) are not disclosed to the model. This is a matched metadata-disclosure
74 test; a single state contains no observed dynamics from which to identify the constants. An evaluation
75 cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s,
76 modalities $\{\text{coords, image, image+coords}\}$ (LLMs only), and prompting $\{\text{direct, chain-of-thought}\}$
77 (LLMs only). Non-LLM models use the coordinate/direct setting.

78 **Out-of-sample protocol.** Learned dynamics models are evaluated off their training data. We
79 generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions
80 (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed
81 by `movement_id`. The cross-family comparison (§5.1) uses horizons $\{1, 10\}$ s, where chaos has
82 begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model. We report no
83 in-sample numbers and quantify the size of the in-sample effect directly (§5.1).

84 3.3 Inputs and grouping

85 Table 3 states what each family receives. Chronos is a time-series model, so we evaluate it in its
86 standard form: a history-conditioned forecaster with a 10 s pre- $t=0$ context window. The LLMs and
87 learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells
88 are equalized by withholding hidden constants and using normal-regime prior constants. A separate
89 asymmetry runs in the opposite direction: the LLMs are large-scale proprietary models, whereas
90 the learned models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B).
91 The behavior of larger physics-informed networks is outside the scope of this study. We separate
92 known-physics methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned
93 dynamics); the former are numerical references.

94 3.4 Metrics

95 The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over
96 links and trajectories. We also track Cartesian bob error, angular-velocity sign match, and energy
97 drift.

98 **Reliability-adjusted error.** A cell is answered only if the model returns a parseable, finite prediction.
99 Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign
100 unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform
101 guess); we also report the strict upper bound π . Answering is thus part of the score.

Verification record. For each evaluation cell we retain the trajectory identifier, model and deployment name, prompt, raw response, parser outcome, prediction, prompt and completion token counts, and score. The released configuration fixes the data-generation solver, held-out seed, horizons, prompts, timeout, and temperature. Two standards apply here. Local methods are reproducible: their checkpoints and seeds regenerate the predictions exactly. Hosted models are not, because a deployment can change or be withdrawn between evaluations, so for them the record is what remains. We therefore release the per-cell responses rather than summary scores alone, which keeps the reported API evaluation auditable after the deployments it measured are gone.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

Disclosure gap. We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, we also run a matched experiment (§5.2) that toggles disclosure on identical trajectories.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing assumptions in the released analysis script.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU in its standard history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels, resampled to up to 512 points, forecasts to the target horizon (prediction length ≤ 64 steps). The comparison uses `chronos-2` in univariate and multivariate modes; the latter forecasts all $2k$ channels jointly. The paired mode comparison uses 20 trajectories per (k, regime) cell. The earlier `chronos-t5-large` was preliminary; Moirai’s `uni2ts` could not be installed in our environment.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation, which Apple’s MPS backend does not run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU. The primary models are trained on nine trajectories, one for each (k, regime) cell (Appendix B). A nested budget study trains fresh Neural ODE and HNN checkpoints with $\{1, 5, 20\}$ trajectories per eligible cell and seeds $\{101, 102, 103\}$. The Neural ODE uses all nine cells; because damping violates Hamiltonian conservation, the HNN budget study uses only the three undamped normal-regime cells. Every checkpoint is evaluated on the same two held-out seeds.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 generates the data and is included as a numerical reference, not an independent competitor.

Baselines. Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the learned models.

On the balanced two-seed subset we also fit SINDy with a degree-two polynomial library and STLSQ, and nonlinear vector autoregression (NVAR) with two delays, degree-two features, and ridge regression. Divergent rollouts are retained as failures and receive the same $\pi/2$ penalty.

Lorenz replication. A secondary Lorenz-63 test compares numerical references, persistence, SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on two five-trajectory held-out seeds. We report training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s; Appendix C gives the full configuration. HNN and LNN are omitted because Lorenz-63 is dissipative and non-Hamiltonian.

5 Results

5.1 Counting failed responses

The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model is evaluated on that grid. The Answered column in Table 4 gives the fraction of cells with valid predictions. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for LLMs they do not include per-call sampling variation.

Table 4 ranks the models on the pendulum under the access conditions in §3.3. The ranking is conditional on those inputs, this training budget, and this system; Section 5.4 tests whether it transfers. The known-equation integrators sit at the top as references (RK4 0.059 at a finer step, symplectic 0.205, Euler 0.405), and every data-driven method is well behind them, closer to the $\pi/2$ random baseline than to the integrators. The lowest-error black-box entry is the frontier LLM kimi (0.538 rad, 99% answered), whose interval excludes the nearest baseline. This establishes its performance for the recorded June 2026 deployment and evaluation conditions. It does not establish a model-family ordering, which the Lorenz test rejects.

Among the trained models the picture is sharper. After equalizing the hidden-regime inputs, nearest-neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972), with the closest intervals overlapping. The stable learned models score about 0.26 rad on their training trajectories and 0.97–1.14 rad on the held-out set, so in-sample evaluation understates their error by about a factor of four.

Two further data-driven methods appear only on the balanced two-seed subset, so we report them in Table 9 rather than in the main comparison: sparse identification (SINDy, 0.883 rad on that subset) falls between the Chronos family and the learned models, and nonlinear vector autoregression (NVAR, 1.248) is among the weakest data-driven methods, near constant velocity.

Effect of missing predictions. grok answers 82% of the grid and lnn 94%. Their ranks fall when missing cells receive $\pi/2$. An answered-cells-only mean omits the cases on which these models failed and gives both a lower reported error.

5.2 Matched disclosure test

On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). Read directly, this looks favorable to withholding constants, but the regimes use different trajectories, and the hidden ones are slower and less divergent, so the comparison cannot speak to parameter recovery. When we instead toggle disclosure on identical trajectories (Table 5), the pooled effect reverses: hiding the constants increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with disclosure helping at $k=1$ and at the 1 s horizon and only a small reverse effect at $k=3$. The unadjusted hidden-regime advantage is thus a difficulty confound.

The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto (am, ad)$ and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. The original single-state probe returned mostly Earth-standard defaults, but we treat those outputs as descriptive and make no system-identification claim from them. The released numerical check verifies both invariances to below 10^{-13} over one-second trajectories.

Table 4: Out-of-sample cross-family comparison on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 generates the data and is included as a numerical reference. Methods new in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	lnn $\dagger$ (full-scale)	1.356	[1.282, 1.430]	94%	learned

Table 5: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

200 5.3 Replication across seeds and training budgets

201 We reran the local pendulum subset on an independent held-out seed with five trajectories per
202 (k , regime) cell; the original seed has 20. Table 9 reports each seed separately and gives their
203 unweighted mean. The selected ordering is unchanged. The numerical references have the lowest
204 errors, Chronos-2 precedes the learned models, and HNN-rollout has the lowest error among the
205 selected learned models. Two seeds do not estimate seed-level uncertainty. The LNN needs a note: its
206 low coverage was a hardware artifact, not model behavior. Its inference solves a Hessian at each step,
207 which exhausted the 8 GB GPU on the longest rollouts and returned those cells as failures. Re-running
208 the LNN on CPU is the same evaluation with the same forecasts; it only recovers the dropped cells,
209 raising primary-seed coverage from 66% to 94%. We use this CPU evaluation for the LNN on both
210 seeds (the second answers all 90 cells, equal-seed rate 97%), so its numbers are produced identically
211 and are comparable across seeds. The cells that still fail are genuine autograd-ODE divergences,
212 scored at $\pi/2$ like any other failure, and the mean stays high at 1.366 rad.

213 **Training-budget check.** With three training initializations and the two held-out seeds, Neural ODE
214 error decreases from 1.090 at one trajectory per cell to 0.968 at five and 0.861 at twenty (Table 6).
215 HNN error on the normal regime falls from 1.269 to 0.974 and then remains 0.981. The HNN and
216 Neural ODE rows cover different regimes and are not an architecture comparison; within each row,

Table 6: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

they show that the nine-trajectory primary result is partly data-limited, while more data alone does not close the gap to known-equation integration.

5.4 Cross-system test: Lorenz-63

We test whether the pendulum conclusions transfer to Lorenz-63, a chaotic system with a polynomial vector field. They do not. Both the identifiability result and the model ordering change.

The identifiability verdict flips. On the pendulum the physical constants are not recoverable from an angle trajectory (§5.2); on Lorenz, sparse identification (SINDy, a degree-two library with STLSQ) recovers the vector field to normalized MAE below 10^{-5} at every horizon. The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is observed in full state, so its parameters are identifiable, whereas the pendulum’s absolute constants are not identifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, which is what the check is meant to expose.

The model ordering also changes. Trained on one seed and evaluated on two held-out seeds, a Lorenz Neural ODE reaches normalized MAE 0.005, 0.055, and 0.258 at horizons 0.1, 0.5, and 1.0 s, below both Chronos-2 modes (0.08–0.11, 0.55–0.57, 0.65–0.66); NVAR is close at 0.1 s (0.012) but drifts to 0.437 by 1.0 s, a persistence baseline trails (0.79 at 1 s), and divergent rollouts are scored at $\pi/2$ as on the pendulum (Table 9). This is the reverse of the pendulum, where the trained models lost to trivial baselines: here a system-trained model beats the zero-shot foundation model. As on the pendulum the families see different inputs (SINDy the library, the Neural ODE target-system training, Chronos a history window), so this is not an architecture race; the cross-system result rejects the pendulum ranking as a general ordering of model families.

The Lorenz test includes the local methods for which the comparison is well defined. It excludes the API LLM grid and the Hamiltonian and Lagrangian networks; a Hamiltonian model does not match the dissipative Lorenz flow. This targeted test is sufficient to falsify transfer of the pendulum ordering and identifiability result.

5.5 Diagnostics

Behavior by horizon. Out-of-sample error by horizon appears in Appendix D (Table 12). At 1 s the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.86–1.26. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.44. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is mid-pack here (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 10 and 11 (Appendix C) report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The changed_hidden regime has the lowest error for almost every model (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding

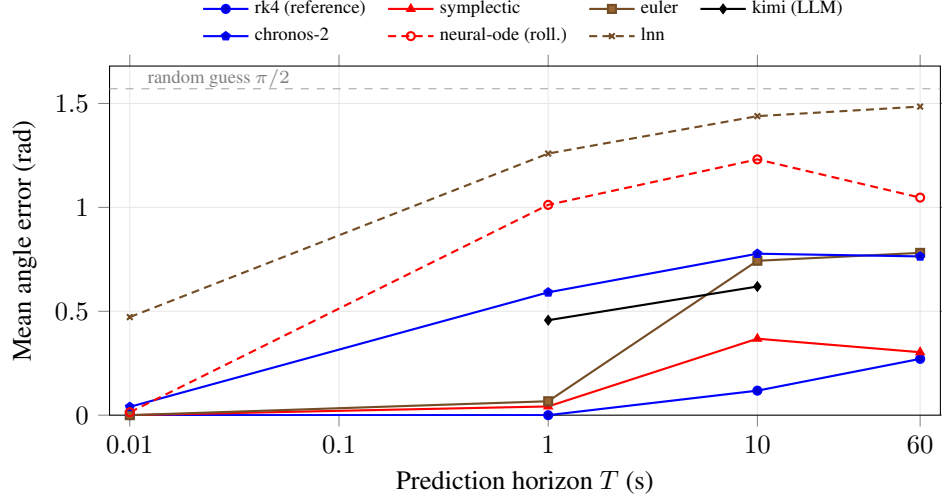


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 7: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19; the previous values favored models given the true constants. The low-gravity, short-rod constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§5.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged. The original bootstrap’s treatment of correlated cells does not determine the reported gaps.

Training-objective check. Table 7 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 8 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged. Changing the integrator therefore does not account for the HNN’s drift. At matched objective, HNN drift is also comparable to Neural ODE

Table 8: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

drift; the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

6 Discussion

The audit changes five conclusions supported by the raw comparison. Held-out evaluation exposes a roughly fourfold learned-model generalization gap; failure accounting penalizes incomplete coverage; a matched intervention reverses the disclosure effect; exact symmetries rule out absolute parameter recovery; and Lorenz-63 changes the model ordering. These are changes to the scientific claim, not to presentation.

More data improves the Neural ODE from 1.090 to 0.861 rad without closing the gap to known-equation integration; HNN improvement plateaus. On Lorenz, SINDy is nearly exact because the vector field lies in its library, while a system-trained Neural ODE beats zero-shot Chronos-2. Performance therefore depends on the system, observations, hypothesis class, data, and access. A scalar leaderboard does not encode those conditions.

The API LLM evaluation is the only billed run. Its per-cell token records give a nominal cost of \$110.38 for 3,960 forecasts, or about \$0.03 each under the June 2026 deployments and pricing. The local methods incur compute cost without a per-query charge.

7 Limitations

The full LLM grid uses one held-out seed; the local replication adds a second, unequal sample. The budget study uses three training initializations and two held-out seeds. Inputs differ by method as listed in Table 3. The Lorenz test uses five trajectories per held-out seed and excludes API LLMs, HNN, and LNN; SINDy’s result depends on a library containing the true polynomial degree. Predictions are endpoints, conservation tests cover only the undamped regime, and foundation-model training corpora are not observable.

8 Conclusion

Correct scores can support wrong scientific conclusions. Here, generalization, failure accounting, matched intervention, identifiability, and cross-system transfer each change a claim supported by the raw leaderboard. Scientific forecasting benchmarks should report those checks, their training budgets, and each method’s information access as part of the result. We chose systems whose ground truth is cheap so that each check could be shown to change a conclusion, but the checks are worth most where verification is expensive; carrying them to systems whose verification budget is the binding

constraint is what this setup was built to support. We release the prompts, outputs, checkpoints, seeds, and compute records needed to reproduce the audit.

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A Prompt templates

The prompts (from `bench/prompts.py`) are reproduced below.

356 **System prompt (direct / no-CoT).**

357 You are a careful physicist predicting the future state of a k -pendulum (a chain of
358 k point masses on massless rods, hinged in a planar chain, swinging under gravity
359 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
360 no code fences. Schema:
361 {"theta": [...k floats..], "omega": [...k floats..]}
362 theta_i is in radians (link i measured from straight-down,
363 positive counter-clockwise)
364 and omega_i is angular velocity in rad/s.

365 **System prompt (chain-of-thought).** Same preamble, then: "Reason step-by-step about the dynamics,
366 then output your final answer. Wrap the final answer in <answer>...</answer> tags containing ONLY a
367 single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}."

368 **User prompt (coordinate modality).**

369 Number of links $k = \{k\}$
370 Physical constants:
371 g (gravity, m/s²): {g}
372 L (rod lengths, m): {L}
373 m (point masses, kg): {m}
374 damping (viscous coefficient on omega): {damping}
375 Initial state at $t=0$:
376 theta_0 (rad): {theta0}
377 omega_0 (rad/s): {omega0}
378 Predict the state at $t = \{T\}$ seconds.
379 Return theta and omega as length- $\{k\}$ JSON arrays.

380 **Hidden-constants regime.** The constants block above is replaced by:

381 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
382 masses, and damping are NOT disclosed. You must infer dynamics from the initial
383 conditions alone.

384 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
385 <answer>...</answer> tags; the first {...} object is JSON-parsed and validated to contain length- k theta
386 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

387 **B Training and compute**

388 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a
389 single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
390 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
391 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
392 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
393 trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
394 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
395 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

396 **Reproducibility records.** The repository holds the code, prompts, trained weights, held-out specifications,
397 and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800),
398 the modality and prompting grid, and the Lorenz sweep. The two large sweeps, the training-budget and balanced-
399 seed studies, are published as release assets. Pendulum training uses seed 42; the primary held-out seed is
400 20260620 (180 trajectories), and the selected local subset also uses seed 20260621 (45 trajectories). LLMs
401 are sampled once at temperature 0. The reported hosted-model numbers recompute from the saved records;
402 re-calling the API is a new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

403 **C Seed and system replication**

404 Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, training seed 42, and
405 held-out seeds 20260620 and 20260621. Each held-out seed contains five trajectories. The Neural ODE is a

406 two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a
407 degree-two polynomial library and STLSQ; NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 9: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of two held-out seeds and then across seeds. Lower is better.

Pendulum model	seed 20260620	seed 20260621	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	0.000	0.000	0.000
neural-ode	0.005	0.055	0.258
nvar	0.012	0.281	0.437
euler	0.029	0.244	0.864
persistence	0.496	1.150	0.790
chronos2-uni	0.106	0.548	0.647
chronos2-multi	0.082	0.568	0.657

408

409 D Breakdowns by link count and regime

410 Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by
411 number of links (Table 10) and by regime (Table 11); models in leaderboard order. Cells a model never produced
412 (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 4.

Table 10: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

Table 11: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity/short-rod constants give slower, less-divergent motion. The unadjusted regime contrast in §5.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 12: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.259	1.439

413 E Additional result tables

414 Full per-model values for the horizon breakdown (§5.5) and the predictability horizon (§5.2). The corresponding
415 summary statistics appear in the main text.

Table 13: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01